

Salt Creek Tiger Beetle Conservation Proposal

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Project Description & Significance:

The Salt Creek Tiger Beetle (SCTB) is globally restricted to the saline-wetlands of Lancaster County, Nebraska. More than 90% of these wetlands have been lost due to hydrological diversion, agriculture, and urban expansion (*OutdoorNebraska*, 2022). The remaining populations fluctuate dramatically, ranging from 150 to 400 adults (US Fish & Wildlife Service, 2021), making the species extremely vulnerable to drought, flooding, and environmental degradation. SCTB depends on groundwater-fed salt flats formed by sodium sulfate discharge from the underlying Pierre Shale. These mudflats provide the correct salinity (>10ppt) required for larval burrow development (Crosier, 2015). When salinity levels drop or vegetation invades this ecosystem, the larvae are unable to survive. This leads to a multiyear population collapse. As the BOS Foundation uses a pre-release island to prepare orangutans for their release into the wild, this project focuses on restoring the microhabitat conditions that SCTB larvae require for their complete development. This project directly advances SAFE program priorities and the USFWS Recovery Plan by restoring hydrology, stabilizing larval habitat, and expanding the breeding areas. The project integrates scientific Monitoring, habitat restoration, and adaptive management, with strong partnerships among landowners, agencies, and AZA institutions.

Goals, Objectives, Action:

The goal of this project is to increase larval survival and expand functional saline mudflat habitat across two priority sites to stabilize SCTB populations. The main objective is to restore the mudflat, maintain salinity levels known to support SCTB larvae, and implement continuous Monitoring to evaluate their success as a species.

The following actions would be taken in order to do so:

- Weekly larval burrow surveys to evaluate emergence, collapse, predation, and moisture levels
- Monthly soil salinity and moisture sampling to ensure that the habitat meets >10ppt (*OutdoorNebraska*, 2022)

- Hydrological restoration by micro-basin excavation and clay plug installation. (BWSR, 2015)
- Targeted reed canary grass and cattail removal to maintain ecosystem structure
- Emergency stabilization following storms or droughts
- Collaboration with landowners and agencies for habitat protection and long-term access

The future goal of this plan is to expand restoration efforts to additional saline wetlands, evaluate habitat corridors, and assess the feasibility of ex situ genetic assurance populations (USFWS, 2021).

Methodology:

This methodology mirrors the structure of the BOS foundation's multi-stage orangutan rehabilitation approach, which involves assessment, intervention, Monitoring, and adaptive management.

Stage 1: Pre-Restoration Assessment

- Soil salinity mapping identifies dilution or freshwater intrusion.
- Moisture profiling determines the saturation duration needed for the larval burrows.
- Larval burrows density mapping will reveal priority microhabitats (USFWS, 2021).
- Hydrological flow-path surveys indicate where to modify the subsurface drainage.

Stage 2: Restoration Work:

- Excavation re-exposes saline mineral soils critical for burrow construction
- Clay plugs prevent rapid desiccation.
- Removal of invasive species (cattails and reeds) prevents shading and soil alteration.
- Emergency stabilization protects larvae following storms.

Stage 3: Monitoring

- Weekly larval monitoring - tracks survival and recruitment trends
- Adult flight season mark surveys estimate population size & movement (USFWS, 2021).

- Quarterly phenology surveys- document vegetation successfully, hydrology, and salt crust formation.

Stage 4: Education & Capacity Building

- Field technicians, interns, and volunteers receive training in wetland ecology and conservation.
- Landowners receive education on saline wetland management & stewardship.
- This method is appropriate because SCTB survival depends on fine-scale environmental conditions, which require precision monitoring.

Timeline:

This project proposal operates on a structured one-year timeline designed to support habitat restoration, Monitoring, and evaluation.

- Months 1-2: Conduct baseline larval burrow mapping, salinity testing, and vegetation surveys. Finalize hydrological restoration plans with partner agencies. Identify high-priority microhabitats for excavation.
- Months 2-4: Implement micro-basin excavation and install clay plugs. Conduct invasive vegetation removal. Begin weekly larval Monitoring.
- Months 3-12: Weekly burrow surveys and monthly salinity/moisture sampling. Quarterly phenology surveys to document hydrological changes. Maintain open mudflat conditions.
- Months 6-12: Conduct biannual evaluation to analyze trends & refine management actions. Prepare midyear and annual reports for project partners and AZA.

This timeline ensures that restoration and Monitoring are coordinated throughout all critical phases of the Beetle's life cycle.

Project Team & Qualifications:

- Project manager: Oversees the entire project and ensures compliance with AZA requirements, coordinates reporting, and manages interagency communication.
- Wetland Restoration Specialist: Directs all hydrological restoration, including micro-basin excavation and invasive vegetation removal—expert in saline wetland structure and soil processes.

- **Monitoring Coordinator:** Develops survey protocols, trains staff, and oversees weekly larval Monitoring, moisture testing, and population analyses.
- **Conservation Biologist: larval ecology:** Provides technical oversight on SCTB microhabitat needs, analyzes larval survival trends, and guides adaptive management.
- **Field technicians** carry out daily field operations, including surveys, soil sampling, vegetative assessments, photo-point Monitoring, and equipment maintenance.
- **Community Liaison:** Works directly with landowners, volunteers, and community partners to support site access, outreach, and conservation awareness.
- **Logistics coordinator:** Manages vehicles, safety compliance, equipment, and scheduling. Ensures field operations are run efficiently

Monitoring & Evaluation:

Monitoring ensures actions are completed. Evaluation determines whether they were effective. This project's Monitoring activities will include:

- Weekly larval burrow activity, density, and condition assessments
- Monthly soil salinity and moisture testing at fixed sampling points
- Vegetation cover assessments to ensure open mudflats
- verification of restoration actions (Excavation & Invasive removal).
- Seasonal hydrology tracking - phenology surveys

Dissemination of Results

Results will be shared through quarterly reports, annual partner meetings, presentations to local conservation groups, and public communication channels. Updated interpretive materials will be provided for the wetland trails, and data will be shared with university researchers for long-term ecological Monitoring and modeling. The internal audience consists of: field technicians, project managers, and all those listed within the project team. The external audience will comprise landowners, local governments, conservation organizations, and the general public.

Permit Status:

- **Section 404 of the Clean Water Act (CWA) Permit:** Required from the **US Army Corps of Engineers (USACE) Omaha District** for any discharge of dredged or fill material into waters of the United States, including wetlands.

(EPA Office of Water, 2025) This is the primary federal permit for this type of activity.

- **Water Quality Certification (Section 401 CWA):** This certification from the [Nebraska Department of Environment and Energy \(NDEE\)](#) is required as part of the federal CWA Section 404 permit process. The NDEE must certify that the activity will not violate state water quality standards (Title 117).
- **Local Zoning and Building Permits:** You must check with the local county (e.g., [Lancaster County](#) or [Saline County](#), depending on the exact location) or city planning/public works department for any specific zoning, grading, or excavation permits.
- **Federal Recovery Permit (Section 10 ESA):** must communicate with USFWS that the project may "take" (harm, harass, collect, etc.) an endangered species or modify its critical habitat, even if for research or conservation purposes. The USFWS issues these permits to enhance the propagation or survival of the species.
- **Landowner access agreement(s):** must obtain legal access onto private state or federal land and permission to alter said land.

Multiyear funding strategy:

Long-term funding will continue through the Nebraska Game & Parks Commission, City of Lincoln Watershed Program, AZA SAFE partner institutions, and regional conservation nonprofits. University grants will support ongoing Monitoring and research.

Past AZA Grant Support:

Past AZA-funded work established SCTB Monitoring methods. These studies confirmed that larval survival correlates with soil salinity and moisture stability (Crosier, 2015). These findings directly shaped the restoration strategies being proposed for this project.

Resources

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